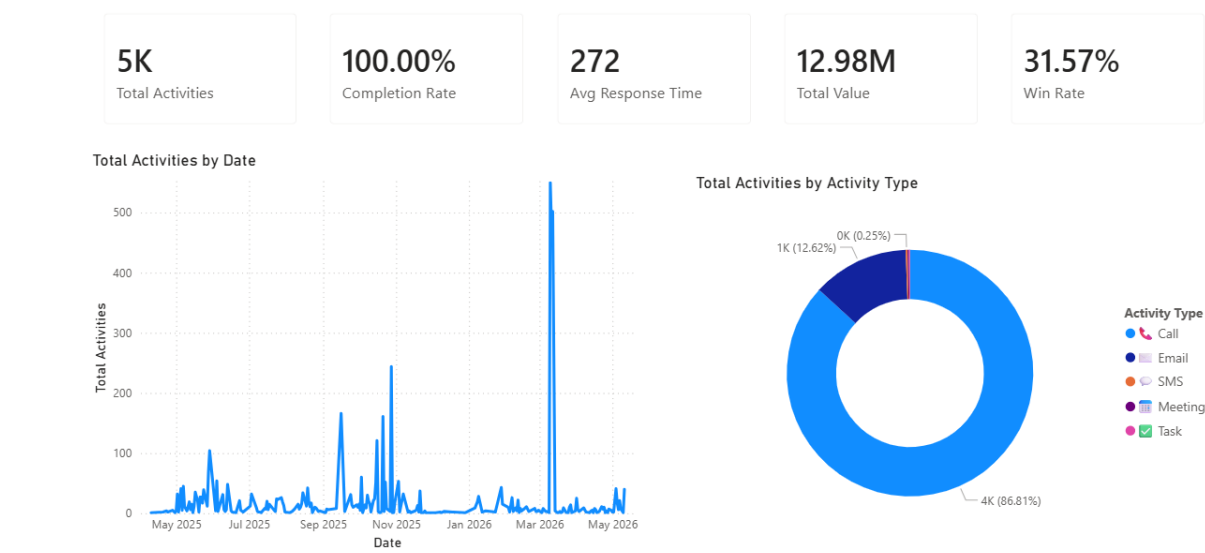


GPM CRM Sales Business Architecture

Michael Ventura



Introduction

This is a proposal for a solution in order to evaluate the process of the data insights from CRM data collected. This document pursuit to compose a technical and functional explanation about how was completed and a possible implementation.

With the CRM Data it has been possible to compose a whole solution for future implementation, which one was delivered in excel spreadsheets, what was in the vision as landscape with this is to complete an automated, enterprise-grade analytics pipeline that transforms your raw data into actionable business intelligence. A complete data platform that:

- Automatically extracts data from your Excel files
- Transforms raw data into analytical models
- Loads into professional databases (PostgreSQL & MySQL)
- Visualizes insights through interactive Power BI dashboards

This document provides the theoretical foundation for the CRM Analytics Pipeline solution. It explains the underlying principles, mathematical models, and academic concepts that inform the architecture and implementation decisions.

The framework covers:

- Data engineering principles
- ETL/ELT architectural patterns
- Database normalization theory
- Dimensional modeling concepts
- Data quality management theory
- Pipeline orchestration patterns
- Visualization best practices
- Mathematical and statistical foundations

The target Audience

- Data Architects
- Data Engineers
- Technical Stakeholders
- Quality Assurance Teams
- Academic Reviewers

Theoretical Framework

Term	Definition	Context in Our Project
ETL	Extract, Transform, Load - Process of extracting data from sources, transforming it, and loading into target systems	Our pipeline extracts from Excel, transforms with dbt, loads to PostgreSQL and MySQL
ELT	Extract, Load, Transform - Data loaded raw first, then transformed within the target database	Used for staging layer where raw data loaded before dbt transformations
Data Pipeline	Automated series of data processing steps	The complete flow from Excel to Power BI
Data Warehouse	Central repository of integrated data from multiple sources	MySQL production database serving as analytics warehouse
Data Mart	Subset of data warehouse focused on specific business area	Our fact_deals table + dimension tables for sales analytics
Staging Area	Intermediate storage where data is prepared before final loading	PostgreSQL staging schema with raw tables
Data Ingestion	Process of importing data from source to storage	read_excel_source.py and load_data_postgres.py
Data Serialization	Converting data structures to storable format	CSV conversion from Excel sheets

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Database Terms

Term	Definition	Context in Our Project
ACID	Atomicity, Consistency, Isolation, Durability - Database transaction properties	Both PostgreSQL and MySQL provide ACID compliance
Normalization	Organizing data to reduce redundancy	3NF achieved through separate dimension tables
Denormalization	Combining tables for query performance	Star schema design for analytics
View	Virtual table based on SELECT query	vw_powerbi_* views for Power BI
Schema	Database structure definition	staging, intermediate, mart schemas

Dimensional Modeling Terms (Kimball)

Term	Definition	Context in Our Project
Fact Table	Table containing measurable business events	fact_deals containing deal metrics
Dimension Table	Table containing descriptive attributes	dim_organization, dim_stage, dim_pipeline
Star Schema	Fact table surrounded by dimension tables	Our core data model

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Term	Definition	Context in Our Project
Grain	Level of detail represented in fact table	One row per deal transaction
Conformed Dimension	Dimension shared across multiple fact tables	dim_date could be shared with future fact tables
Degenerate Dimension	Dimension stored in fact table without its own table	deal_id serves this purpose

dbt-Specific Terms

Term	Definition	Context in Our Project
Model	SELECT statement that creates a dataset	staging_deals.sql, fact_deals.sql
Macro	Reusable SQL logic	calculate_cycle_time(), clean_lost_reason()
Hook	SQL executed before/after model builds	post-hook for grants
Seed	CSV file loaded as a table	lost_reason_mapping.csv
Test	Validation query that returns failing rows	assert_deal_value_positive.sql
Generic Test	Built-in test (unique, not_null, accepted_values)	Applied to multiple columns
Singular Test	Custom SQL test	Custom business logic tests

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Term	Definition	Context in Our Project
Documentation	Auto-generated docs from .yml files	dbt docs serve
Lineage Graph	DAG showing model dependencies	dbt documentation site
Run	Executing models to build/update tables	dbt run
Compile	Generating SQL from Jinja templates	dbt compile
Freshness	Checking source data timeliness	source freshness tests
Exposure	Declaring downstream uses of models	Power BI dashboard exposure

Data Quality Terms

Term	Definition	Context in Our Project
Data Profiling	Analyzing data structure and content	Initial EDA on Excel file
Data Validation	Checking data against rules	dbt tests, schema validation
Data Cleansing	Correcting or removing invalid data	lost_reason standardization, null handling
Data Deduplication	Removing duplicate records	Is_deleted flag, primary keys

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Term	Definition	Context in Our Project
Null Handling	Managing missing values	COALESCE, IS NULL checks
Outlier Detection	Identifying anomalous values	Statistical methods on value distribution
Referential Integrity	Ensuring foreign key relationships	relationships generic test
Business Rule Validation	Checking domain-specific rules	value ≥ 0 , valid status values
Data Freshness	How recently data was updated	ingested_at timestamps
Anomaly Detection	Identifying unexpected patterns	Monitoring win_rate changes
Data Quality Score	Composite metric of quality dimensions	Future enhancement
Data Observability	Monitoring data pipeline health	Logging, alerting

Business Intelligence Terms

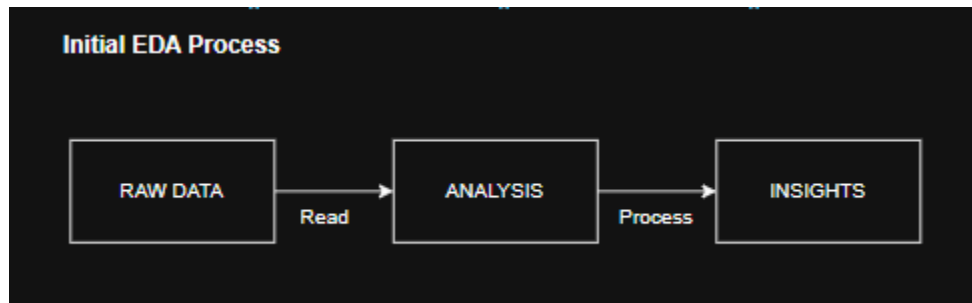
Term	Definition	Context in Our Project
KPI	Key Performance Indicator - Measurable value	Win Rate, Pipeline Value, Cycle Time
Dashboard	Visual display of KPIs and metrics	Power BI dashboard with 5 pages

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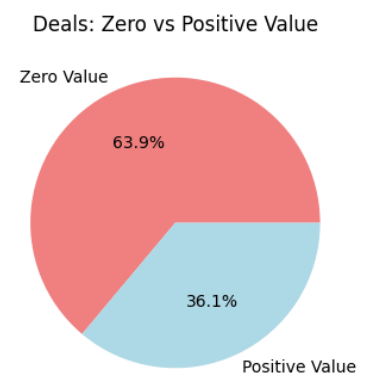
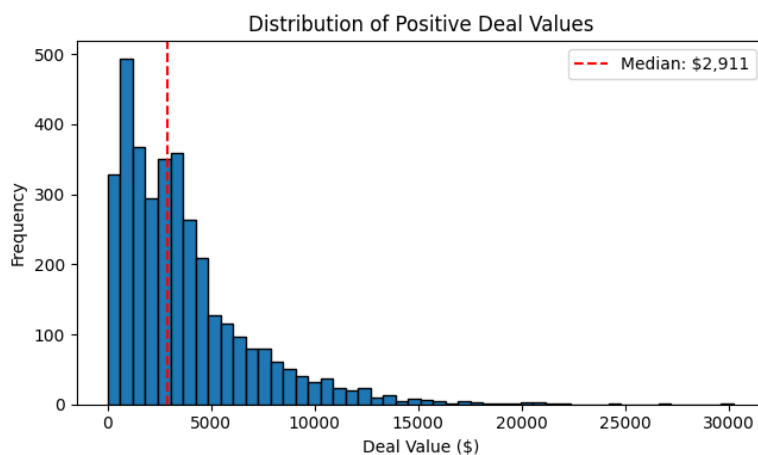
Term	Definition	Context in Our Project
Drill-Through	Navigating from summary to detail	Click on KPI to see detailed data
Drill-Down	Expanding hierarchy to see sub-levels	Organization → deal details
Filter	Restricting data based on criteria	Date range, pipeline type, organization
Slicer	Visual filter control in Power BI	Interactive filtering controls
Measure	DAX formula for calculations	Total Revenue, Win Rate DAX
Calculated Column	Row-level computation in table	value_bucket, activity_level
DAX	Data Analysis Expressions - Power BI formula language	Custom metrics and calculations
M Language	Power Query formula language	Data transformation in Power BI
Import Mode	Loading data into Power BI memory	Current approach (9,756 rows fits)
Aggregation	Pre-summarized data for performance	kpi_daily_summary tabl

EDA Process

It was first initialized with an exploratory data analysis where we could have on first hand initial insight for the pipeline process. We could find duplicated rows for activities and deals, but related with names and subjects, besides of this there is a good data integrity in relation with IDs.

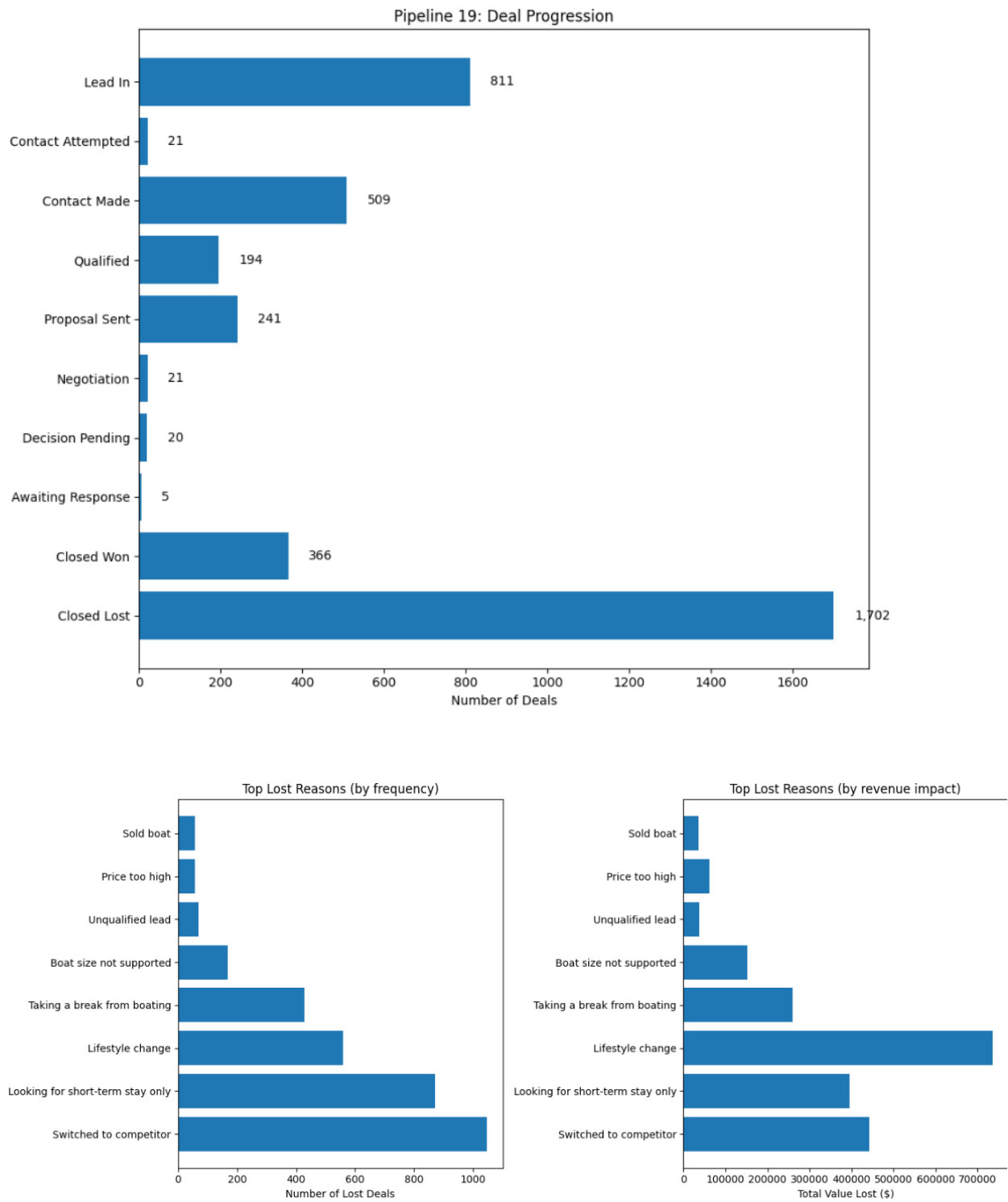


There are most of zero-values deals those are not won, in addition of this won deals have significantly higher value that lost/open, indicating that larger deals are more likely to close.



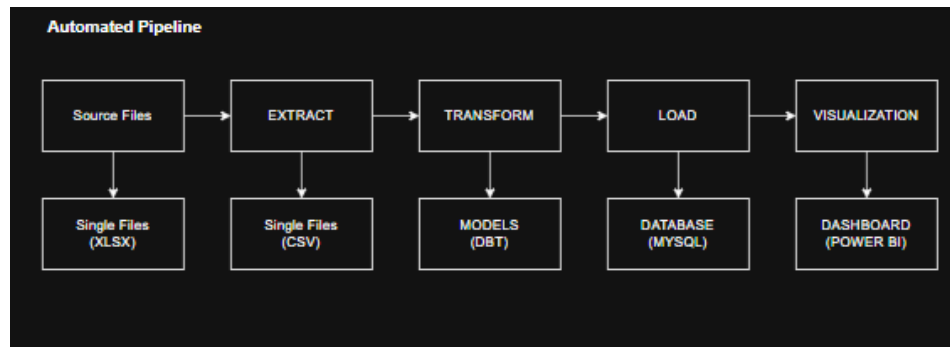
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The funnel reveals a major data quality or tracking issue, the conversion and drop-off percentages are mathematically impossible in several stages. This win rate data is much cleaner than the funnel data, and reveals a stark contrast in performance across pipelines.

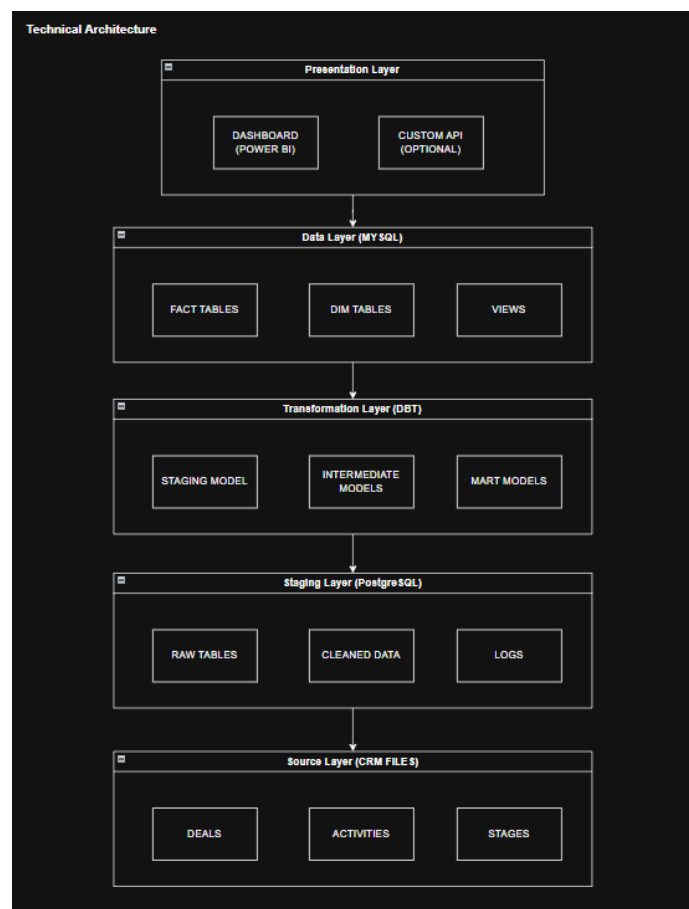


Architecture

At this stage we evaluate to implement an automated scheduled ETL that can valid data with more accuracy, having historical tracking for compliance, interactive dashboard to take faster decisions, focus in scalability using docker and better approach with data lineage.



If we organize the hole process into a graphical focus, segmenting into layers we could have this scenario as landscape

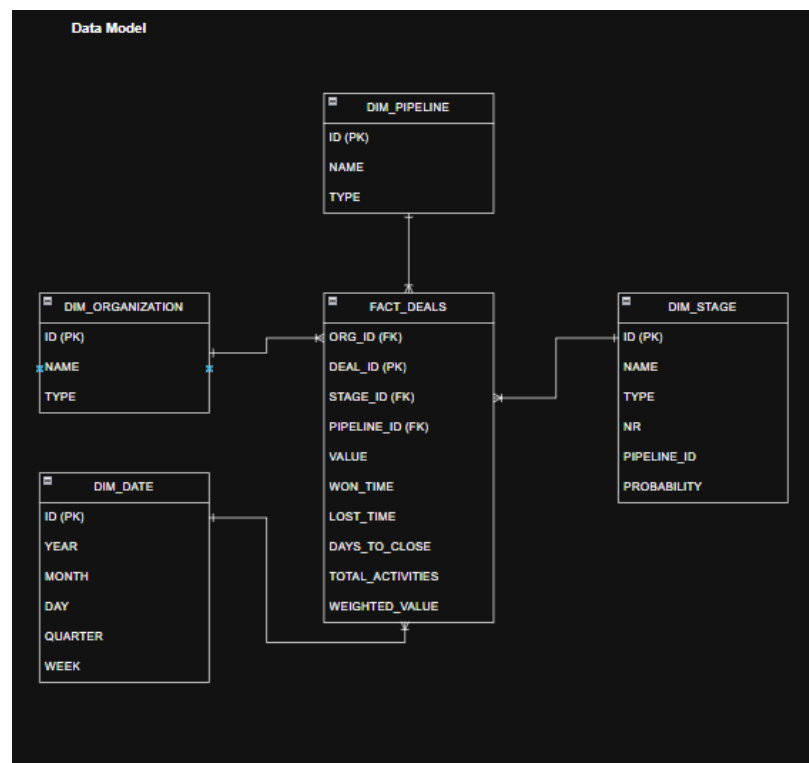


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The designed architecture for this solution uses this composition:



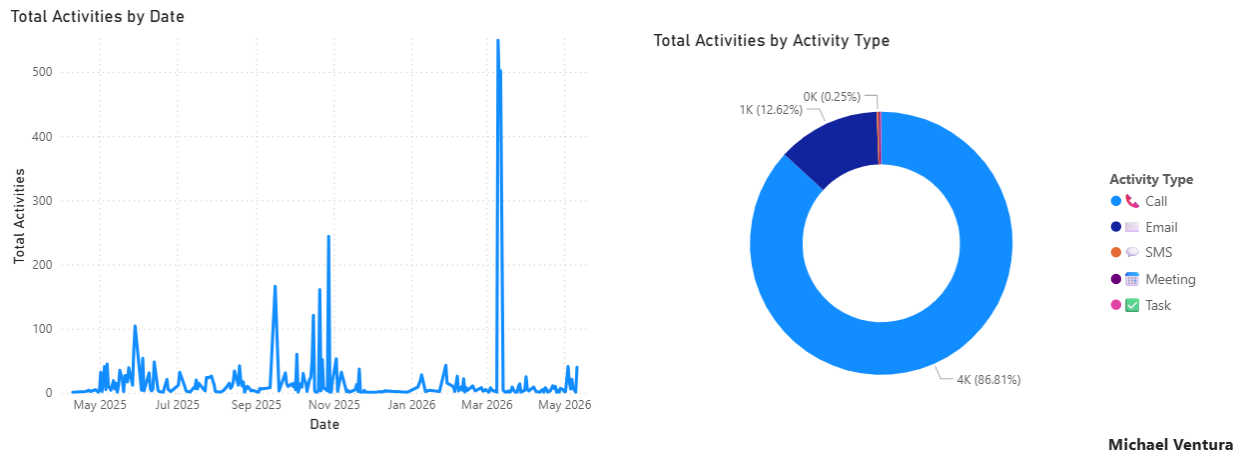
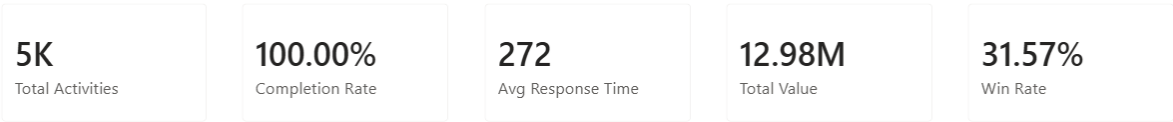
The data model oriented to STAR schema, orbiting over fact tables some dimensions



Insights

After completing the whole process of ETL, and populated the data as incremental focus, then we can evaluate different opportunities for dashboards

CRM Summary

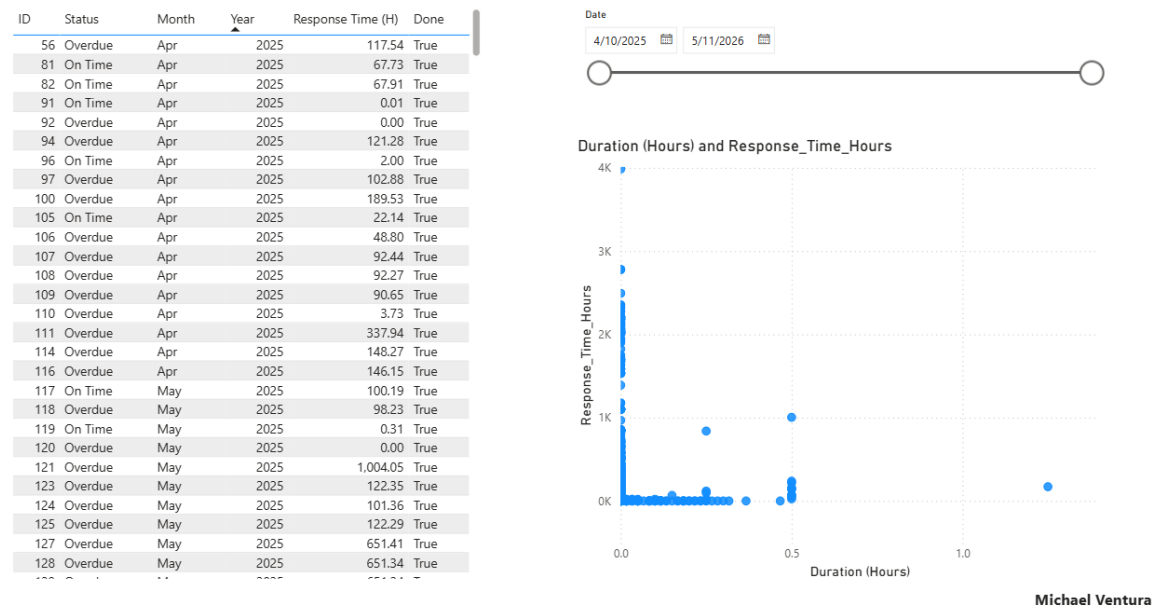


Summary of full process into dimensions and fact tables, with some KPI, exploring the total of activities, the completion rate, average of response, and total value and win rate.

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Exploring insights trying to leave what matters into the visualization

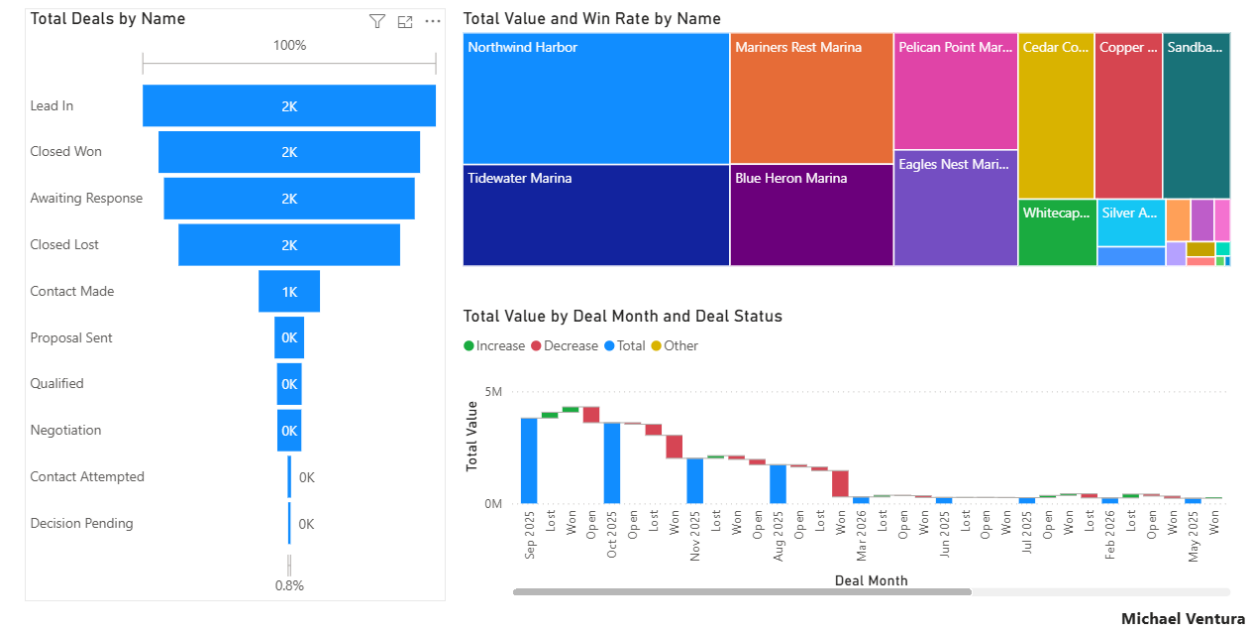
CRM Activity Details



In full picture we can show the activities and duration slicing by date

At the end another page with sales, focused on segments, using funnels and treemaps to evaluate quicker with visual references

CRM Sales Pipeline



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